

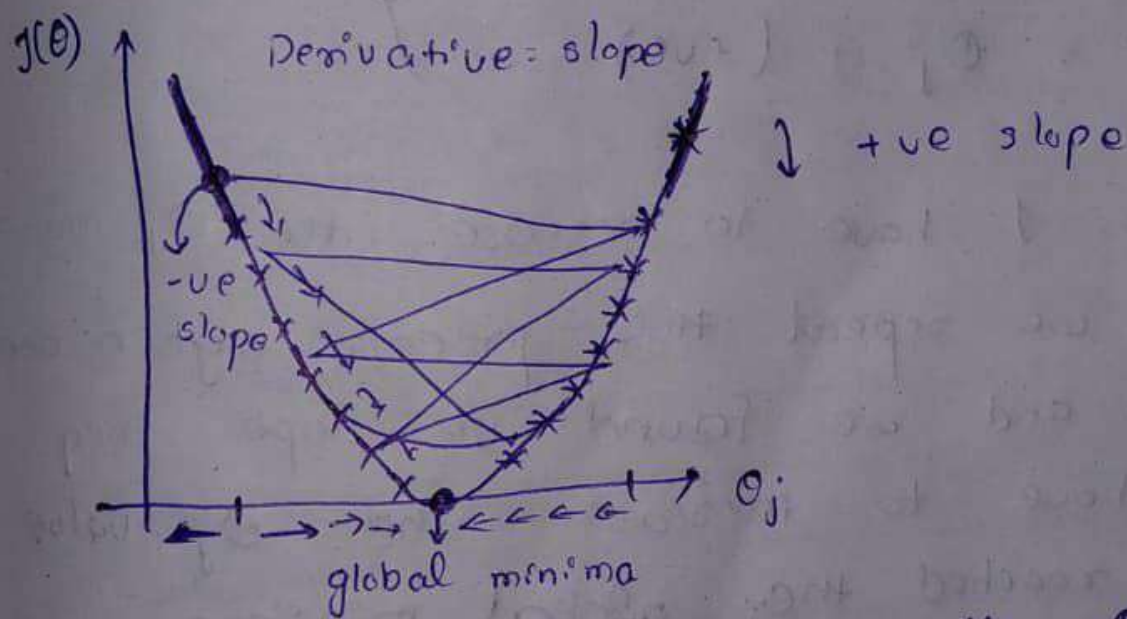
Convergence Algorithm

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↓ optimize the changes of θ_i values

Repeat until Convergence

$$\theta_j = \theta_j - \alpha \left[\frac{\partial}{\partial \theta_j} J(\theta_j) \right] \rightarrow -ve$$



Note: This equation change the θ_i value much more efficiently

Make or draw a tangent from the point and if the right side is downward then it is -ve slope.

if you want to find the derivative

$$\boxed{\frac{\partial}{\partial \theta_j} J(\theta_j)}$$

→ This value will be -ve.

if -ve ↗ learning rate

$$\left\{ \begin{array}{l} \theta_j = \theta_j - \alpha (-ve) \\ \theta_j = \theta_j + (+ve) \end{array} \right\}$$

Means I have to increase the θ_j value.
So, if we repeat this process again and again and we found -ve slope then we have to increase the θ_j value so I reached the global minima.

$$\theta_j^0 = \theta_j - \alpha \left[\frac{\partial J(\theta_j)}{\partial \theta_j} \right] \rightarrow \text{true}$$

true slope when right side of slope facing upward then we have to decrease the θ_j value and repeat the process again and again to reach the global minima.

$$\begin{aligned} \theta_j^0 &= \theta_j - \alpha (+ve) \\ &= \theta_j - (-ve) \end{aligned}$$

α = learning rate

let $\alpha = 0.001$ it is a small value

learning rate control the speed at which convergence happen.

If value is very very small then take more time to converge and if it is big then continuously jump here and there and then never converge.

$\alpha = 0.001 \rightarrow$ best practice